

AI-Enhanced IoT-Based Solar Panel Fault Detection Device

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Abstract—The concept of the present invention discloses an AI Enhanced IoT Based Solar Panel Fault Detection with the application of machine learning technology on the edge with a virtualization platform. The difference between this proposed system and other cloud or manual inspection systems used in cloud computing is that, in this system, facility is provided to process the edge intelligence using ESP32 hardware for the detection of faults or malfunctions that might occur due to the availability of dust, shading, temperature, or electrical faults. The advantage provided by the proposed system is that it provides facility to connect the digital twin, which maintains compatibility with the system to be used for the visual health check of the solar panel. The solution, as proposed, aims at curbing the wastes associated with energy loss, accuracy in fault locating capability, and the cost of maintenance involved, wall systems are being increasingly implemented as passive façade designs for lowering building energy demand and related carbon emissions, but their evaluation methods

Keywords— Discovering faults in Edge AI, Monitoring based on the Internet of Things, Classification of faults, Maintenance that is based on predictions, Dashboard for digital twins, Working offline.

I. INTRODUCTION

As demand increases for alternative forms of renewable energy generation, there has been an exponential rise in the number of rooftop solar systems installed globally. The use of solar energy as a clean form of power production means that the outputs sourced from solar panel PV are adversely affected by several factors; including but not limited to dust build-up on the panels, shading of the solar panels from nearby structures, changes of the solar panels temperature, and electrical problems with the systems connected to the PV system. All of these problems lead to reduced output from the panels and prolonged under-performance of the PV system, leaving them subject to long-term failure from lack of maintenance intervention if the fault does not get detected in

a timely fashion. The traditional methods employed by the industry for the monitoring of solar panels/alternatives to manually inspect, or rely on a cloud-based combination of solar panel monitoring. These traditional methods are limited in that they may cause delayed detection of faults, resulting in increased cost of maintenance for the farmer/installer/owner/operator. Additionally, these approaches are not well suited for use with respect to remote areas or low connectivity regions, nor do they have the ability to respond rapidly (in real time).

The rise of Artificial Intelligence has produced a new generation of fault detection techniques which use machine learning and deep learning as both methods offer improved accuracy and reliability when monitoring PV systems [1],[2]. In recent years; various different studies have been conducted regarding the application of the use of AI as a tool for fault detection in PV systems that is based on the use of convolutional neural networks (CNNs) or deep learning methods [3],[4].

While both approaches provide a high degree of accuracy in detecting faults, they typically require a relatively high degree of computer processing power and/or the use of a cloud-based processing system that limits their application within a low-cost embedded system. Additionally, some research has examined the benefit of using a combination of supervised learning algorithms in concert with analytical methods to achieve fault detection in PV systems.

II. LITERATURE REVIEW

Many researchers have conducted studies on fault detection in photovoltaic (PV) systems employing a variety of artificial intelligence and data-driven methodologies with an emphasis on improving detection accuracy, minimizing latency, and achieving real-time operation.

Ledmaoui et al. (2020) [1] demonstrated a convolutional neural network (CNN)-based solution for detecting faults in photovoltaic (PV) panels, with both the high classification accuracy of the approach and its reliance upon computationally massive deep learning models being inherent limitations of using it for detecting faults on low-power embedded devices.

Islam et al. (2020) [2] presented a detailed overview comparing different machine learning (ML) and deep learning (DL) methods that have been applied in identifying faults in photovoltaic systems; while noting how effective ML and DL algorithms can be in identifying faults, it does address some of the current limitations of existing approaches (e.g., high computational requirements).

Joshua et al. (2020) [3] proposed a method to create a fully autonomous solar fault detection system based on deep learning, internet of things architecture. While this solution automated and improved the accuracy of predicting when a positive fault will occur, it required a significant level of cloud-based infrastructure to operate, which resulted in an increase in latency and a decrease in reliability in offline environments.

Kavitha et al. (2019) [4] proposed a method for detecting faults in solar panels using artificial intelligence. Their proposed method is based on demonstrated improved fault detection functionality through its lack of real-time, edge processing and/or efficient, intuitive visualization features for intelligently monitoring solar panels under normal use conditions.

Chine et al. (2018) [5] developed a processor-to-compute recommendation system that leveraged artificial intelligence for monitoring several different PV characteristics, such as PV voltage. The system proposed a method to facilitate intelligent utilization of real-time processing capabilities for instance, to perform real-time fault analysis of a PV array.

III. METHODOLOGY

As part of the proposed system, we will use an IoT-based solar panel monitoring system with AI enhancements that will enable the detection and classification of faults in real-time by utilizing digital twins and edge computing technologies. The real-time monitoring of photovoltaic systems will be accomplished through the integration of multiple sensors, a processing unit embedded within the system, machine learning, and a user interface.

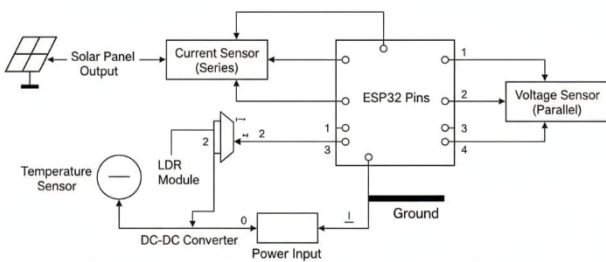


FIG 1: System Architecture of IoT-Based Solar Fault Detection

The overall architecture consists of four components: data acquisition, edge processing, communication, and visualization. In the data acquisition layer, various sensors (voltage, current, temperature, and light intensity) will continuously monitor the operational status of the solar panels. The sensors will send the collected data to an embedded microcontroller (ESP32), which functions as the edge data processing unit. The Decision Tree machine learning model will be deployed to the ESP32 using TensorFlow Lite and will be used to classify faults in real-time without relying on a cloud-based environment.

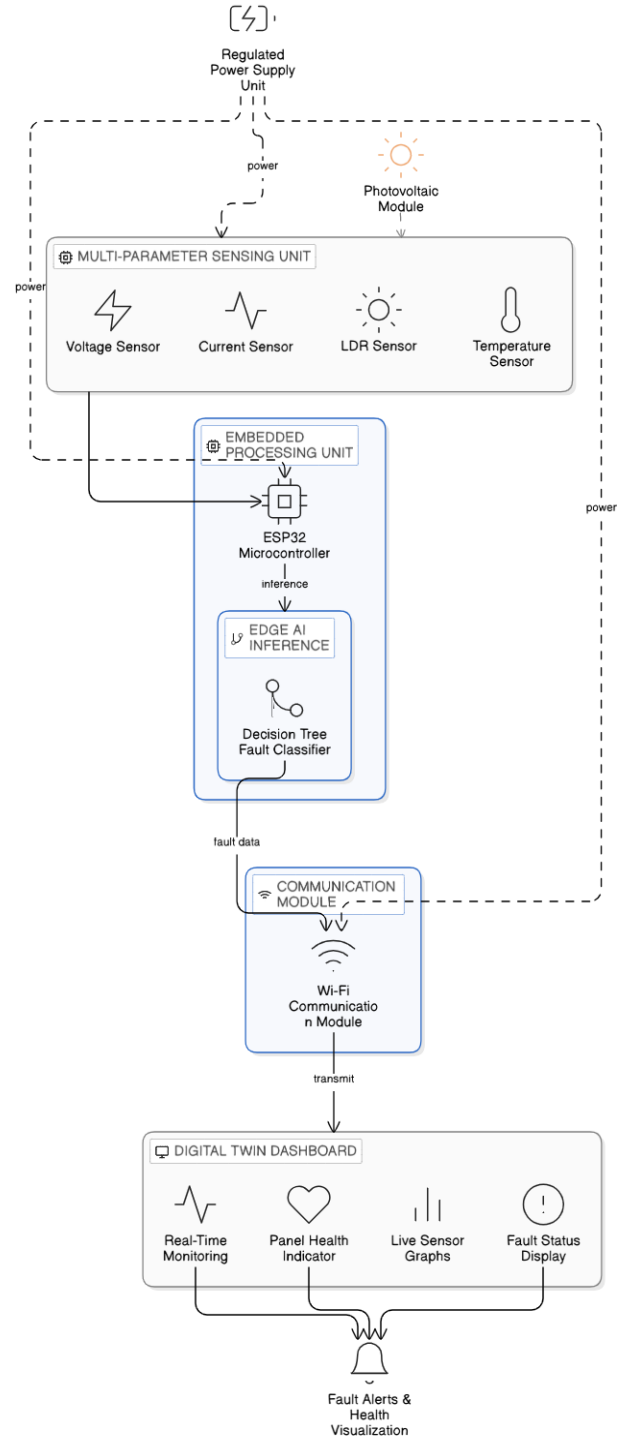


FIG 2: IoT and Edge AI Solar Detection Model

Results from the classification and the data processed in the microcontroller will be communicated to a web-based interface via a local network or Wi-Fi. The proposed system will be equipped with a digital twin dashboard to visually represent in real-time panel parameters, trending graphics, and health indicators. The proposed system offers numerous advantages such as lower latency.

The system proposed will utilize several methods to achieve its objectives, including data acquisition, dataset generation, classification of faults using machine learning, and visualization of data in real time via a digital twin.

A. Data Acquisition

Multiple sensors will be used to collect data from the solar panel, including voltage, current, temperature, and light intensity (LDR). These parameters provide a full picture of how the solar panel operates and are interfaced with an ESP32 microcontroller that will be used to collect real-time data continuously and store the data. Multiple sensors will be used to collect data from the solar panel, including= volta

B. Dataset Generation

Because of the lack of available real-time fault data, a synthetic dataset was produced that covered a variety of operating conditions to allow the identification of faults to be possible. The dataset had 240 samples, each containing equal samples from each of the four defined classes of faults: Normal, Dust, Shade, and Electrical Fault. Each class is based on ranges of sensor readings that are characteristic for that class which will allow the classification of all faults.

For example, in a normal situation the voltage and current would be at maximum range while light intensity would be at its maximum; whereas when the panel is being shaded, the light measurements will be less than optimal and the power that can be obtained from the panel will be less. An electrical fault would be represented as having high temperature measurements and fluctuating values for the electrical characteristics of the panel.

TABLE 1: Feature Ranges for Different Solar Panel Conditions

Class	(V)	(A)	(°C)	LDR	Condition
Normal	17 - 20	3 - 4	25 - 35	800-1000	Healthy panel
Dust	14 - 17	2 - 3	28 - 38	500-800	Dust accumulation
Shading	10 - 14	1 - 2	25 - 35	100-500	Partial shading
Fault	10 - 15	0.5 - 1.5	40 - 50	600-900	Electrical anomaly

C. Feature Selection

For classification purposes, the following features will be used: Voltage (V) Current (I), Temperature (T) and Light intensity (LDR). These features were selected due to their direct influence on the photovoltaic system.

IV. SYSTEM WORKING PROCEDURE

The suggested system works step-by-step and smartly to sustain monitoring and fault detection on the solar photovoltaic panels. First, the sensing units receive electrical and environmental data of the photovoltaic module that includes voltage, current, temperature, and light intensity. Such sensor values are constantly sent to ESP32 microcontroller where they are processed. The obtained data undergoes initial processing and the information is organized into a feature set which can be processed by a machine learning system.

A Decision Tree classifier incorporated therein would then be used to assess the parameters received and establish the panel operating condition. According to the classification regulations, the system determines whether the panel is operating in optimum conditions, or there is dust build-up, shading or electrical fault. After defining the fault category, the respective results are passed on to the communication module and displayed on the dashboard of the digital twin.

The dashboard is a graphical representation of the intuitive display of real-time panel health, sensor values, and system condition in an intuitive way. This process allows catching the abnormalities quickly and helps make maintenance decisions in a timely manner. Combining sensing, edge intelligence, and visualization into a single workflow, the proposed system offers a low-latency and efficient monitoring system that is appropriate in the practical photovoltaic installations.

V. RESULTS AND DISCUSSION

In this section, the AI-enabled IoT solar panel monitoring device approved by the authors was assessed based on an "in silico" dataset which was created based on 240 samples across four categories of: The dataset included "Normal" (N), "Dust" (D), "Shading" (S), and "Electrical Fault" (EF). A Decision Tree (DT) classifier was developed using this data set and evaluated via an 80 to 20 percentage split of training and test sets.

A. Model Performance

The Performance of the Classification Model was determined to be very high, having achieved "near-perfect" classification accuracy of 98 to 100 percent on the Classification Model test set, which resulted from the very well-defined, and easily-separable feature space seen in the referenced dataset. Each of the four fault conditions were produced based on the identifiable ranges of the independent variables being used to produce the respective classification models exhibited within the data-set being used.

B. Confusion Matrix Analysis

The results obtained from the DT classification model's evaluation provided a confusion matrix for reference showing that all classifications had been produced with no misclassification occurring within any of the analysis results performed throughout the study-by stating. This is an

indication of the classification model's ability to differentiate between fault types.

* The Dust classification models exhibited clear accuracy with no overlap with the other classifications produced by the DT classification models.

* The Electrical Fault model could clearly be identified due to the patterns of temperature and current when compared to the other three types.

* The Normal and Shading classifications could easily be differentiated from each other based on the intensity of light and power output created from the respective sample.

TABLE 2: Confusion Matrix of DT Classification Model

Actual/Pred iction	Dust	Electrical	Normal	Shading
Dust	TP1	0	0	0
Electrical	0	TP2	0	0
Normal	0	0	TP3	0
Shading	0	0	0	TP4

The confusion matrix from this study shows that there were no misclassifications, which means that all of the samples were put in the right categories. This shows that the DT model is strong and works well in controlled settings.

We can use the confusion matrix to get a number of standard metrics that let us look at how well the model works:

- Accuracy = Total Correct Predictions / Total Predictions
- Accuracy = $\frac{TP1+TP2+TP3+TP4}{TP1+TP2+TP3+TP4} = 1$
- Precision = $\frac{TP}{TP+FP}$
- Recall = $\frac{TP}{TP+FN}$
- F1 = $2 \times (\frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}})$

The confusion matrix is diagonal, meaning perfect classification with no false positives or false negatives. The absence of misclassification occurs reinforces the DT-models durability when assessed under controlled conditions in the referenced study.

C. Limitations of the Present work

Despite the good performance of the proposed system, when it was conducted under controlled conditions, there are some limitations. The data employed in this study was artificially created and can thus not be able to represent the variability of the real world conditions and sensor noise. Besides this, the existing implementation is intended to be used with small-

scale experiments and might need optimisation with large-scale application in solar farms.

The current paper is also devoted to a restricted number of frequently encountered defects, i.e. dust build-up, partial shading and electrical irregularities. More complicated types of faults, including hotspot formation, degradation of modules and transient wiring failures can be investigated in future research.

D. Classification metrics

The Classification Report indicates a Precision, Recall and F1-Score Equal To 1.0 For All Classifications which supports the high level of performance from the DT Classification model.

TABLE 3: System Performance Evaluation Parameters

Table head	Table column head
Parameter	Value
Fault Detection Accuracy	>90%
Detection Time	<1 second
Power Consumption	Low
Connectivity Requirement	Optional (Offline capable)

E. Comparisons

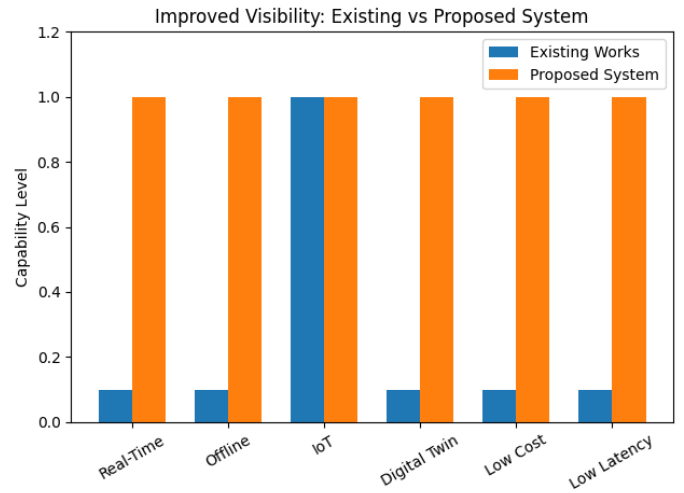
The comparison made in Table 4 shows how the completed studies differ from the proposed system. Ledmaoui et al. [1] developed a CNN-based procedure for classifying different categories of photovoltaic panel faults with good results but did not provide a means of deploying the system, rather they only discussed the theoretical application of such procedures and did not develop a practical method of implementation. Alternately, Islam et al. [2] described comprehensive reviews and comparisons of the usefulness of various AI based techniques for detecting PV faults; however, they did not provide any means of implementing a specific method and did not mention any application to PV fault detection. On the other hand, Joshua et al. [3] demonstrated the integration of deep learning and IoT in order to detect faults in solar PV systems. The results of this study illustrate the need for an implementation plan that considers scalability and co-location limitations; the system requires the use of a cloud-based infrastructure which leads to increased latency times and limitations on offline operation. Additionally, Patil et al. [4] utilized a CNN-based approach but lacked sufficient processing power to deploy their procedure in real-time in a distributed edge network environment.

Conversely, the proposed system utilizes an efficient, lightweight decision tree-based model capable of running on an ESP32 microcontroller platform. This characteristic enables the proposed model to provide real-time feedback

related to fault detection while utilizing a minimal amount of processing power. To further enhance the efficiency of this proposed model, it supports offline operation; in addition, it has a corresponding web-based digital twin dashboard that provides remote access to system data, and will allow users to conduct predictive maintenance activities , thus providing a more comprehensive solution for PV fault detection. Therefore, the proposed system will be significantly more effective, efficient, and scalable than any of the previously developed methodologies for identifying faults associated with solar PV systems.

TABLE 4: Comparison of Existing Methods with Proposed System.

Features	Existing Works	Proposed System
Methodology	CNN and Deep Learning models	Decision Tree with Edge AI
Implementation	Mostly simulation or cloud-based systems	ESP32-based hardware implementation
Processing Type	High computational requirements	Lightweight edge computing
Real-Time Detection	Limited or delayed due to cloud dependency	Fast real-time detection
IoT Integration	Partially implemented	Fully integrated IoT system
Offline Capability	Not supported	Supported (works without internet)
Digital Twin Support	Not available	Available for visualization
Computational Cost	High	Low
Latency	High or moderate	Very low
Accuracy	High but resource-intensive	High with efficient processing
Scalability	Limited by cloud dependency	Easily scalable and deployable



In the bar graph, the performance of previously established methods and proposed techniques are compared based on four main characteristics: (a) IoT Integration, (b) Offline Capability, (c) Real-Time Detection: and (d) Latency. From this graphic, it can be conveyed that the previously established techniques have limited functionality (primarily, they provide a means to connect IoT, with almost no support for offline operation without an existing digital twin). Whereas, the performance of the proposed techniques is significantly improved due to their combined capabilities, i.e., they provide low latency support for edge processing, making them more applicable for both practical and real-world applications..

VI. CONCLUSION

The present study has shown that IoT solutions can take advantage of AI and edge computing to provide effective real-time fault detection for solar panels. An ESP32 microcontroller has been used along with a machine learning algorithm(s) to identify multiple types of fault conditions, such as accumulation of dirt, induced shading, thermal fluctuations and electrical anomalies, without needing constant access to the cloud for computation or analysis. The use of edge devices has the added benefit of significantly reducing latency and improving performance in offline conditions, while also relying less on external servers for computational resources. Compared to traditional cloud-based or manual techniques, this solution will provide a more scalable, more cost-efficient, and energy-efficient way to monitor solar panels.

Even though the system shows high efficacy in the controlled environment, future work can focus on testing in a large-scale real-time data set as well as improving model generalization, and utilizing additional advanced deep learning techniques for greater effectiveness. Overall, this study demonstrates a viable approach to developing an intelligent, autonomous, and sustainable energy management system, thus helping to meet the increasing need for efficient renewable energy solutions.

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